MONITORING & CONTROLLING COSTS USING BEST PMP PRACTICES & PROJECT MANAGEMENT GUIDELINES

1. PROJECT PROGRESS

1. Percent Complete

$$\% \text{ Complete} = \frac{\text{Actual Quantity}}{\text{Total Quantity}} \times 100$$

2. Physical Progress

$$\text{Physical Progress} = \frac{\sum(\text{Weight}_i \times \text{Progress}_i)}{\sum \text{Weight}_i}$$

3. Remaining Progress

$$\text{Remaining} = 100 - \% \text{ Complete}$$

2. EARNED VALUE MANAGEMENT (EVM)

1. Planned Value (PV)

$$PV = \text{Budget} \times \text{Planned \% Complete}$$

2. Earned Value (EV)

$$EV = \text{Budget} \times \text{Actual \% Complete}$$

3. Actual Cost (AC)

$$AC = \sum \text{Actual Expenditure}$$

4. Budget at Completion (BAC)

$$BAC = \sum \text{Budget}$$

3. VARIANCE ANALYSIS

1. Cost Variance (CV)

$$CV = EV - AC$$

2. Schedule Variance (SV)

$$SV = EV - PV$$

3. Cost Variance %

$$CV\% = \frac{CV}{EV} \times 100$$

4. Schedule Variance %

$$SV\% = \frac{SV}{PV} \times 100$$

4. PERFORMANCE INDICES

1. Cost Performance Index (CPI)

$$CPI = \frac{EV}{AC}$$

2. Schedule Performance Index (SPI)

$$SPI = \frac{EV}{PV}$$

3. Critical Ratio (CR)

$$CR = CPI \times SPI$$

5. FORECASTING

1. Estimate at Completion (EAC)

$$EAC = \frac{BAC}{CPI}$$

(or) $EAC = AC + (BAC - EV)$

2. Estimate to Complete (ETC)

$$ETC = EAC - AC$$

3. Variance at Completion (VAC)

$$VAC = BAC - EAC$$

4. To Complete Performance Index (TCPI)

$$TCPI = \frac{BAC - EV}{BAC - AC}$$

6. COST CONTROL

1. Cost Growth

$$\text{Cost Growth} = \frac{\text{Final} - \text{Budget}}{\text{Budget}} \times 100$$

2. Unit Cost

$$\text{Unit Cost} = \frac{\text{Total Cost}}{\text{Quantity}}$$

3. Cost Efficiency

$$\text{Efficiency} = \frac{\text{Budgeted Cost}}{\text{Actual Cost}}$$

4. Budget Utilization

$$\text{Budget Used} = \frac{\text{Actual}}{\text{Budget}} \times 100$$

7. CASH FLOW

1. Net Cash Flow

$$\text{Cash}_{\text{Net}} = \text{Income} - \text{Expenses}$$

2. Cumulative Cash Flow

$$\text{Cumulative} = \sum \text{Monthly Cash Flow}$$

3. Burn Rate

$$\text{Burn Rate} = \frac{\text{Actual Cost}}{\text{Time}}$$

4. Monthly Spending Rate

$$\text{Rate} = \frac{\text{Monthly Cost}}{\text{Working Days}}$$

8. PROCUREMENT

1. Commitment

$$\text{Commitment} = \text{PO} + \text{Subcontracts}$$

2. Outstanding Commitment

$$\text{Outstanding} = \text{Commitment} - \text{Paid}$$

3. Retention Balance

$$\text{Retention} = \text{Certified Value} \times \text{Retention \%}$$

4. Advance Recovery

$$\text{Recovery} = \text{Invoice} \times \text{Recovery \%}$$

9. PRODUCTIVITY

1. Labour Productivity (LP)

$$LP = \frac{\text{Output}}{\text{Labour Hours}}$$

2. Equipment Productivity (EP)

$$EP = \frac{\text{Output}}{\text{Machine Hours}}$$

3. Resource Utilization (RU)

$$RU = \frac{\text{Actual Hours}}{\text{Available Hours}} \times 100$$

4. Crew Efficiency

$$\text{Efficiency} = \frac{\text{Earned Hours}}{\text{Actual Hours}} \times 100$$

10. PLANNING& SCHEDULING

1. Total Float (TF)

$$TF = LS - ES = LF - EF$$

2. Free Float (FF)

$$FF = ES_{next} - EF$$

3. Activity Duration

$$\text{Duration} = \text{Finish} - \text{Start}$$

4. Remaining Duration

$$\text{Remaining} = \text{Original} - \text{Elapsed}$$

11. RISK MANAGEMENT

1. Expected Monetary Value (EMV)

$$EMV = \text{Probability} \times \text{Impact}$$

2. Contingency

$$\text{Contingency} = \sum EMV$$

3. Risk Exposure

$$\text{Exposure} = \text{Probability} \times \text{Consequence}$$

12. CONTRACTS & FINANCIAL KPIs

A. CONTRACTS

1. Liquidated Damages (LD)
$LD = \text{Rate} \times \text{Delay Days}$
2. Retention
$\text{Retention} = \text{Invoice} \times \text{Retention \%}$
3. Price Escalation (PE)
$PE = \text{Contract Value} \times \text{Escalation Index}$
4. Change Order %
$CO\% = \frac{\text{Variation}}{\text{Original Contract}} \times 100$
5. Contract Completion %
$\text{Completion \%} = \frac{\text{Certified Value}}{\text{Contract Value}} \times 100$

B. FINANCIAL KPIs

1. Gross Profit (GP)
$GP = \text{Revenue} - \text{Cost}$
2. Gross Margin (GM)
$GM = \frac{\text{Revenue} - \text{Cost}}{\text{Revenue}} \times 100$
3. Profit Margin (PM)
$PM = \frac{\text{Profit}}{\text{Revenue}} \times 100$
4. Return on Investment (ROI)
$ROI = \frac{\text{Profit}}{\text{Investment}} \times 100$

COMMON ABBREVIATIONS

PV	= Planned Value	GPI	= Schedule Performance
EV	= Earned Value	CV	= Schedule Variance
AC	= Actual Cost	SV	= Schedule Variance
BAC	= Budget at Completion	TF	= Total Float
EAC	= Estimate at Completion	FF	= Free Float
ETC	= Estimate to Complete	PO	= Purchase Order
VAC	= Variance at Completion		

NOTES

- All costs should be in same currency.
- Time unit must be consistent (days/weeks/months).
- Use baseline (BAC & schedule) approved for comparison.
- Keep progress measurement method consistent.
- Update actuals regularly for reliable forecasts.

KEY PERFORMANCE AT A GLANCE

CPI > 1 → Under Budget
CPI = 1 → On Budget
CPI < 1 → Over Budget

SPI > 1 → Ahead of Schedule
SPI = 1 → On Schedule
SPI < 1 → Behind Schedule

Step No.		Cost-Control Method	What to Do	Why It Matters	Target / Control Measure
1.		Establish the Cost Baseline	Approve the time-phased project budget before execution begins.	Provides the official reference for measuring cost performance.	100% approved before work starts.
2.		Develop a Cost Breakdown Structure	Divide the total budget into work packages, activities, resources, materials, and accounts.	Makes cost ownership and tracking easier.	Every cost linked to a CBS code.
3.		Link Costs to the Work Breakdown Structure	Assign each cost item to the corresponding WBS activity.	Connects scope, schedule, and budget control.	100% cost-to-WBS alignment.
4.		Define Cost Control Accounts	Create control accounts for departments, contractors, phases, and major deliverables.	Improves accountability and performance measurement.	One responsible owner per account.
5.		Track Budget Versus Actual Cost	Compare planned expenditure with actual expenditure at regular intervals.	Identifies overspending before it becomes critical.	Cost variance within ±5%.
6.		Calculate Cost Variance	Use the formula CV = EV − AC.	Shows whether completed work is under or over budget.	CV ≥ 0.
7.		Monitor Cost Performance Index	Use the formula CPI = EV ÷ AC.	Measures the cost efficiency of completed work.	CPI ≥ 1.00.
8.		Update the Cost Forecast	Calculate the expected total cost at project completion.	Provides early warning of a potential budget overrun.	EAC ≤ approved budget.
9.		Calculate Estimate to Complete	Use the formula ETC = EAC − AC.	Shows the remaining funding required to finish the project.	Realistic and regularly updated.
10.		Monitor Committed Costs	Track purchase orders, contracts, approved invoices, and pending obligations.	Prevents hidden financial commitments from being overlooked.	Actual plus committed cost within budget.
11.		Control Scope Changes	Evaluate the cost impact of every requested change before approval.	Prevents uncontrolled scope growth and budget leakage.	No work without approved change authorization.
12.		Manage Contingency Reserves	Release contingency funds only for approved and documented risks.	Protects reserves from unnecessary consumption.	Reserve usage aligned with risk exposure.
13.		Review Procurement Costs	Compare supplier quotations, contract rates, and market prices.	Reduces purchasing costs and improves commercial value.	Procurement variance within approved limits.
14.		Control Material Waste	Measure ordered, consumed, damaged, returned, and unused materials.	Reduces waste, theft, rework, and storage costs.	Waste percentage below project threshold.
15.		Monitor Labor Productivity	Compare actual labor hours and output with the approved productivity baseline.	Identifies inefficient working practices and overtime exposure.	Productivity equal to or above baseline.
16.		Improve Equipment Utilization	Track productive hours, idle time, maintenance, and rental duration.	Prevents unnecessary equipment and rental expenses.	Utilization ≥ 85%.
17.		Reduce Rework Costs	Record the cost of defects, rejected work, repairs, and repeated activities.	Protects both the project budget and schedule.	Rework cost ≤ 5% of contract value.
18.		Validate Invoices and Progress Claims	Confirm quantities, rates, approvals, and completed work before payment.	Prevents duplicate payments and inaccurate billing.	100% invoice verification.
19.		Conduct Monthly Cost Reviews	Review variances, commitments, cash flow, risks, forecasts, and corrective actions.	Keeps management informed and supports timely decisions.	One formal cost review every month.
20.		Implement Corrective Action	Assign owners, deadlines, and recovery measures when costs deviate from the baseline.	Turns cost information into measurable financial improvement.	Corrective action started within seven days.

Essential Project Cost-Control Formulas

1. Cost Variance: $CV = EV - AC$
2. Cost Performance Index: $CPI = EV \div AC$
3. Estimate at Completion: $EAC = BAC \div CPI$
4. Estimate to Complete: $ETC = EAC - AC$
5. Variance at Completion: $VAC = BAC - EAC$
6. To-Complete Performance Index: $TCPI = \text{Remaining Work} \div \text{Remaining Budget}$

Drw Purpose Code :-	R01- Educational
Status of Approval :-	01
Dimentions as Mentioned	NA
Drawing Scale :-	NTS
Engg Code :-	99
Backup Info Code :-	04
Angle of Projection	NA
Sheet Size	A3

Statutory Note:
This document is prepared by rPlan and is the sole proprietory of the company.Its purpose is educational and no references within this document are for Engineering or Design corelation with the applicable codes. For any clarifications beyond the intended purpose viewers can always correspond to rPlan via its company website.This drw is not for duplication without company permission!

Drw electronic code

rPlan
AT 39, MERMAID 1 CHS LTD., SEC-11, CBD-BELAPUR, NAVI MUMBAI - 400614

Legal Owner

QR CODE

NA

Description:	PROJECT CONTROLS Cost Monitoring & Controlling on Projects
Drawn By:	DFT-01
Drawn Date	20260618
Checked Approved by	rPlan-01
Checked Approved Date	20260910
Drawing Number	As Barcoded
Sheet	1 of 1
Rev	00